

// Problem 2

```
#include <iostream>
#include <string>

using namespace std;

// GameCharacter Class which represents a playable charecter in an RPG
// encapsulation, constructors, and state management
class GameCharacter {
private:
    string name;
    int health;
    int level;
    int strength;

public:
    // Default Constructor
    GameCharacter() {
        name = "Freelancer";
        health = 100;
        level = 1;
        strength = 10;
    }

    // Allows for naming and HP
    GameCharacter(string charName, int hp, int str) {
        name = charName;
        health = hp;
        strength = str;
        level = 1;
    }

    // Measure for taking damage starting at 0
    void receiveDamage(int points) {
        health -= points;
        if (health < 0) health = 0;
        cout << name << " took " << points << " damage!" << endl;
    }

    // Measure to restore health
    void heal(int points) {
        health += points;
        if (health > 100) health = 100; // Assuming 100 is max for level 1
        cout << name << " restored " << points << " health." << endl;
    }

    // Increases level and stats
    void levelUp() {
        level++;
        strength += 5;
        cout << "*** " << name << " reached Level " << level << "! ***" <<
```

```
endl;
}

void showStatus() {
    cout << "\n--- Character Sheet ---" << endl;
    cout << "Name:      " << name << endl;
    cout << "Level:     " << level << endl;
    cout << "Health:    " << health << endl;
    cout << "Strength:  " << strength << endl;
    cout << "-----\n" << endl;
}

};

int main() {
    // Instantiate a new character
    GameCharacter player("Valerius", 90, 15);

    // Demonstrate object behavior
    player.showStatus();

    player.receiveDamage(25);
    player.heal(10);
    player.levelUp();

    player.showStatus();

    return 0;
}
```